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Directly Converting Syngas to Linear α-Olefins over Core−Shell Fe_3O_4@MnO_2 Catalysts

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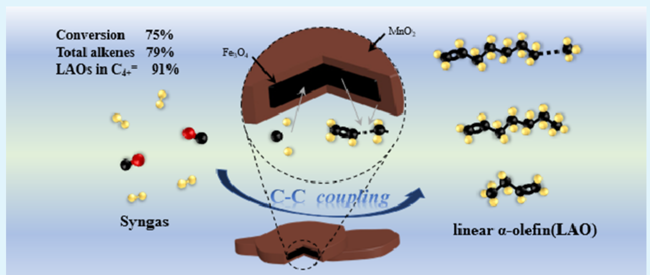
Supporting Information

ABSTRACT: Converting syngas to value-added chemicals via Fischer−Tropsch synthesis has attracted much attention, whereas the direct hydrogenation of CO to heavy olefins, especially linear α-olefins (LAOs), remains a challenge. In this study, we designed a core−shell Fe_3O_4@MnO_2 catalyst to realize the direct conversion of syngas to LAOs with high efficiency. This catalyst exhibited a high selectivity of 79.60% for total alkenes and 64.95% for C_4^+ alkenes, 91% of which are LAOs, at a CO conversion of approximately 75%. Promotion of the electron transfer from MnO_2 to Fe_3O_4 inside the core−shell Fe_3O_4@MnO_2 catalyst facilitated the dissociative adsorption of CO molecules on Fe_3O_4 and the spillover of H atoms onto the MnO_2, which enhanced C−C coupling, weakened the hydrogenation activity of the catalyst, and improved the production of LAOs. A superior stability over 100 h was observed, demonstrating the promising potential of this catalyst for industrial applications.



KEYWORDS: *Fischer−Tropsch synthesis, linear α-olefins, Fe_3O_4@MnO_2, C−C coupling, syngas*

1. INTRODUCTION

Fischer−Tropsch synthesis (FTS) is capable of producing clean liquid fuels or other chemicals from syngas derived from biomass, coal, and natural gas and has received increasing interest for both academic and industrial applications because of environmental pollution and the shortage of transportation fuels.[1−5] Because of the very broad product distribution, which ranges from C_1 to C_{50} hydrocarbons and follows the Anderson−Schulz−Flory (ASF) model, obtaining specific products via FTS in industrial applications is restrained.[6,7] Recently, there has been a significant progress in developing novel catalyst systems that break the ASF distribution by altering the electronic and structural properties of active metal sites.[4,8,9]

Light olefins (C_{2-4} olefins), which can be obtained by the direct conversion of syngas in the FTS process, are widely utilized as important chemical feedstocks for producing polymers and/or drugs.[10,11] In recent years, Jiao et al.[12] developed a $ZnZrO_x$/SAPO-34 bifunctional catalyst that presented a high C_{2-4} olefin selectivity (∼80%) and low CH_4 selectivity (∼2%). Cheng et al.[13] also designed a composite oxide−zeolite of ZnO−ZrO_2/SAPO-34 and achieved a C_{2-4} olefin selectivity of 70% at 10% CO conversion. Because of the thermodynamic and kinetic differences between these two reactions, including methanol synthesis and the conversion of methanol to olefins, precise adjustments are very difficult from an engineering standpoint. Directly converting syngas to light olefins over iron-based catalysts, as reported by Galvis and de Jong,[6] has gained considerable attention. Alkali promoters are typically added to iron-based catalysts to enhance CO adsorption and weaken H_2 adsorption, promoting the formation of light olefins.[14]

In addition to light olefins, heavy olefins, especially linear α-olefins (LAOs) containing a terminal C═C bond, are desired chemical intermediates for a large number of high value-added chemicals, such as polymerization monomers,[15] surfactants,[16] and premium synthetic lubricants,[17,18] which are currently produced by ethylene oligomerization from petroleum resources.[19,20] However, over the past decade, less attention has been paid to the creation of a method to produce heavy alkenes via the direct conversion of syngas (Table S1) than to the development of a process to convert methanol to light olefins. Linghu et al.[21] prepared a Co/SiO_2 catalyst with a 41.4% LAO selectivity. Subsequently, Zheng et al.[22] reported a Mo-modified Co/SiO_2 catalyst with a total α-olefin selectivity of 74.2%. Recently, Fe-based catalysts have attracted increasing attention because of their low cost, poison resistance, excellent water-gas shift (WGS) ability, and high selectivity for olefins.[23]

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Gao et al.[24] reported a traditional iron-based catalyst that presented a greater than 69% α-olefin selectivity. The results of Zhai et al.[25] suggested that a Na-modified Fe_5C_2 catalyst exhibited a high selectivity for C_5^+ alkenes, particularly LAOs. However, the difficulty in controlling the C−C coupling and the hydrogenation ability of active sites makes it challenging to design highly efficient catalysts for selectively converting syngas to LAOs.

In particular, manganese as an effective promoter has enabled electronic tuning of iron by enhancing electron transfer during FTS, which enhances CO chemisorption and inhibits hydrogenative chemisorption, promoting the formation of olefins.[26] Liu et al.[1] reported that the Mn-modified Fe_3O_4 catalyst improved the selectivity for light olefins to 60.1%. The results of Zhang et al.[27] demonstrated that Fe_2O_3@MnO_2 spindles exhibited a high selectivity of 66.6% for C_5^+ hydrocarbons. The excellent electronic tuning ability of the Mn promoter provides a new path to optimize LAOs selectivity; however, use of the Mn-modified Fe catalyst for the direct transformation of syngas to LAOs has been rarely reported to date.

In this study, we synthesized a core−shell Fe_3O_4@MnO_2 catalyst via a one-pot hydrothermal method. This catalyst presented an outstanding total alkene selectivity of 79.60% and C_4^+ alkene selectivity of 64.95%, of which LAOs accounted for 91.04%, at a CO conversion of 75.47%. Moreover, N_2 adsorption−desorption analysis, scanning electron microscopy (SEM), transmission electron microscopy (TEM), X-ray photoelectron spectroscopy (XPS), powder X-ray diffraction (XRD), hydrogen temperature-programmed reduction (H_2-TPR), thermogravimetric analysis (TGA), and laser Raman spectroscopy (LRS) were combined to characterize the core−shell structure of the catalyst and reveal the electronic modulation of the Mn promoter.

2. EXPERIMENTAL SECTION

2.1. Catalyst Preparation. The single Fe_3O_4 catalyst was synthesized by a one-pot hydrothermal method. Analytical grade poly(vinylpyrrolidone) (PVP), ferrous sulfate ($FeSO_4{\cdot}7H_2O$), potassium permanganate ($KMnO_4$), and sodium hydroxide (NaOH) were purchased from Sinopharm Chemical Reagent Co., Ltd. (Shanghai, China). First, $FeSO_4{\cdot}7H_2O$ (13.9 g) and PVP (10.0 g) were together dissolved in 1 L of water. After being kept at 90 °C for 1 h, NaOH (40 mL) was added to the solution to generate a green suspension, which became a black suspension after constant stirring for 4 h. Then, the sample was collected by centrifugation and washed with deionized water and ethanol several times.

Fe_3O_4@MnO_2 catalysts were synthesized by a one-pot hydrothermal method.[28] After obtaining the green suspension mentioned above, $KMnO_4$ aqueous solutions of different concentrations were added dropwise to the suspension to produce dark brown precipitates. After being aged for 12 h, the sample was collected by centrifugation, followed by washing and drying. The synthesized single Fe_3O_4 catalyst was denoted as Mn-free, whereas the prepared Fe_3O_4@MnO_2 catalysts were denoted as Mn-X, where X is the content of Mn in molar percentage. The content (mol %) of Mn varied from 10 to 50%. The catalysts were sieved to 40−60 mesh without being pelleted.

2.2. Catalyst Characterization. The Brunauer−Emmett−Teller (BET) specific surface area was measured by a Micromeritics ASAP 2020 at −196 °C. Before testing, all catalysts were degassed under vacuum at 120 °C for 3 h. The detailed element content in our prepared catalysts was obtained by inductively coupled plasma (ICP) spectrometry using a Varian 720-OES spectrometer. All catalysts were dissolved in aqua regia at 90 °C for 4 h. The XRD spectra were obtained on a Bruker AXS-D8 Advance (Germany) diffractometer equipped with Co Kα (λ = 1.78 Å) radiation (35 kV, 40 mA) with a scanning angle (2θ) range of 10°−90°. XPS measurements were conducted on a VG Systems (MultiLab 2000) equipped with an Al Kα (1486.6 eV) quartz monochromator source. All peaks were corrected by setting the C 1s peak of 284.6 eV as the reference. LRS of our catalysts was performed on a RM-1000 confocal Raman microscope.

H_2-TPR measurements were carried out in a quartz-tube fixed-bed microreactor. Fifty milligrams of the sample was in situ pretreated with a high purity N_2 stream (30 mL·min^{-1}) at 150 °C to remove the residual water and other contaminants. After being exposed to N_2 for 1 h, the catalyst was switched to a 5% H_2/95% N_2 flow exposure and heated to 800 °C at a rate of 10 °C·min^{-1}. The H_2 contents of the tail gas were continuously recorded by a thermal conductivity detector (TCD).

SEM images were acquired with a Quanta F250 scanning electron microscope. Scanning TEM and energy-dispersive X-ray (EDX) mapping measurements were performed with a Philips CM200 high-resolution TEM (HRTEM) to obtain further insights into the surface morphology and structure. TGA was carried out in air atmosphere with a thermal analyzer (STA 409 PC) using a 10 mg sample and a 10 °C·min^{-1} heating rate.

2.3. Catalytic Evaluation. The catalytic reactions were conducted in a stainless fixed-bed tubular reactor (inner diameter = 10 mm, length = 40 cm). Typically, the prepared catalyst (around 0.5 g, 40−60 mesh) diluted with quartz sand (0.5 g, purchased from Tianjin Nankai Chemical Plant, China) was reduced in situ by H_2 (26 mL·min^{-1}, 0.1 MPa) at 350 °C for 10 h. The catalytic test was conducted in syngas (5% N_2/47.5% CO/47.5% H_2) under the desired reaction conditions. The operation conditions were as follows: temperature of 280−360 °C, gas hourly space velocity (GHSV) of 3000−12 000 h^{-1}, and pressure of 1.0−4.0 MPa.

The FTS products were analyzed by FULI GC 97 gas chromatographs. The tail gas was analyzed online by flame ionization detector with an RB-PLOT Al_2O_3 capillary column and TCD with a 5A MolSieve packed column and Porapak Q column. The collected liquid products were analyzed by FID with a HP-5 column. The α-olefins and other products were analyzed by 450 gas chromatography (GC)−320 mass spectrometry (MS) (a typical GC−MS result is shown in Figure S1). The product selectivity was analyzed on a carbon basis, and the time-on-stream was 30 h.

The CO conversion is obtained by

$$\text{CO conversion} = \frac{CO_{in} - CO_{out}}{CO_{in}} \times 100\%$$

CO_{in}: the volume of CO at the inflow and CO_{out}: the volume of CO at the outflow. The CO conversion observed in the blank tests (in the absence of catalyst, only by the effect of the temperature) is zero.

The CO_2 selectivity is obtained by

$$CO_2 \text{ selectivity} = \frac{CO_{2\ out}}{CO_{in} - CO_{out}} \times 100\%$$

$CO_{2\ out}$: the volume of CO_2 at the outflow.

The hydrocarbon selectivity (product distribution) is calculated by

$$C_iH_m \text{ selectivity} = \frac{i \times C_iH_{m\ out}}{\sum_{i=1}^{n} i \times C_iH_{m\ out}} \times 100\%$$

The selectivity to a hydrocarbon does not take into account the formation of CO_2.

$C_iH_{m\ out}$: the mole number of the hydrocarbon with i carbons at the outflow.

FTY, which is the converted CO moles per gram in unit time, can be obtained by

$$\text{FTY (mol}\cdot g_{Fe}^{-1}\cdot s^{-1}) = \frac{Q_{in}\nu_{CO}x_{CO}}{V_m m_{Fe}}$$

Q_{in}: the inlet total volume flow rate; ν_{CO}: the molar percentage of CO in syngas; x_{CO}: the CO conversion; V_m: the gas molar volume under

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standard conditions, V_m = 22 400 mL/mol; and m_{Fe}: the mass of Fe in the catalyst.

The value of chain growth probability (α) determines the hydrocarbon distribution produced by FTS and is defined by

$$\alpha = r_p/(r_p + r_t)$$

where r_p and r_t mean the chain growth rate and the chain termination rate, respectively.

In our experiment, the chain growth probability can be fitted and calculated by

$$\ln\left(\frac{W_n}{n}\right) = n \ln \alpha + \ln \frac{(1-\alpha)^2}{\alpha}$$

W_n: the mass fraction of the hydrocarbons with n carbons.

3. RESULTS AND DISCUSSION

3.1. Characterization of Fe_3O_4@MnO_2. The morphological and structural characteristics of the prepared Fe_3O_4@MnO_2 catalysts are shown in SEM and TEM images in Figure 1. For the synthesized single Fe_3O_4 catalyst, a nanosphere with a size of approximately 26 nm is observed in Figure 1A. The phase structures of the as-prepared catalysts are further analyzed by XRD (Figure 2). The XRD pattern shows diffraction peaks at 2θ of 30.0°, 35.4°, 43.0°, 53.4°, 57.0°, and 62.5°, which are ascribed to the face-centered cubic Fe_3O_4 (JCPDS card number: 19-0629),[29] confirming the formation of Fe_3O_4 nanoparticles. When the Mn species is introduced, the spherical particles gradually become flattened, accompanied by the formation of a roughened surface and blurred edges (Figure 1B,C). More nanoplates appear with the increasing Mn content (Figure 1D,E), indicating that the incorporation of the Mn species in Fe_3O_4@MnO_2 facilitates the formation of a flattened morphology. According to the XRD results (Figure 2), the gradual addition of Mn species to Fe_3O_4@MnO_2 leads to the appearance of a peak located at 2θ = 40.5°, which may be because of the formation of amorphous MnO_2. With the addition of the Mn promoter, the intensity of the Fe_3O_4 peaks decreases, whereas that of MnO_2 peaks slowly increases, suggesting that the introduction of the Mn species promotes the dispersion of Fe_3O_4 nanoparticles and decreases the iron crystal size. Moreover, the XRD pattern presents no obvious diffraction peaks of the Fe/Mn solid solution, which means that the Mn species does not substitute for the octahedral site of Fe_3O_4.

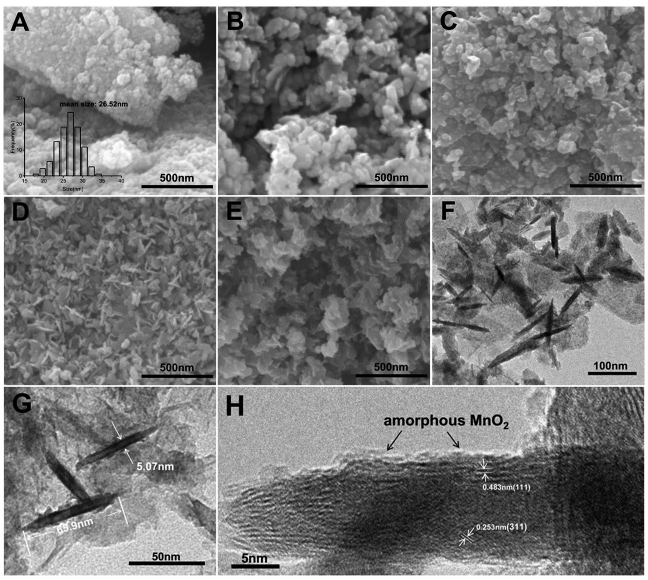

Figure 1. SEM image of fresh catalysts (A) Mn-free, (B) Mn-10, (C) Mn-18, (D) Mn-33, and (E) Mn-50 and TEM image (F,G) and HRTEM image (H) of fresh Mn-50.

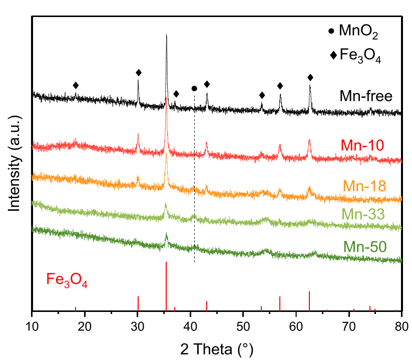

Figure 2. XRD patterns of the fresh Fe_3O_4@MnO_2 catalyst.

The typical Mn-50 catalyst (Fe/Mn molar ratio of 1:1) is further analyzed by TEM. As shown in Figure 1F, plate-shaped architectures with an average edge length of approximately 66 nm and an average thickness of 5 nm (Figure 1G) can be clearly observed in the TEM image, which is consistent with the SEM results. Interestingly, the morphology of our samples has a high degree of similarity with that of Zhang et al.,[28] indicating that we have prepared the same rod-shaped core—shell architectures as theirs through the basically same experimental steps. According to the corresponding HRTEM image (Figure 1H), lattice spacings of 0.483 and 0.253 nm are obtained in Mn-50, which are assigned to the (111) and (311) planes of the Fe_3O_4 phase, respectively. In addition, the amorphous MnO_2 coating on the surface of the Fe_3O_4 nanoplates can be observed on the edge of the lattice fringe. No lattice spacing of the Fe/Mn solid solution appears inside the nanoplates, which is consistent with the XRD results. These results suggest that the Mn species added to the Fe_3O_4 nanoparticles mainly covers the surface, producing a typical Fe_3O_4@MnO_2 core—shell structure. Moreover, the corresponding EDX mapping of fresh Mn-50 (Figure 3) shows the distribution of Fe and Mn. The surface layer is rich in Mn, whereas Fe is mainly located inside the catalyst, further confirming the coverage of MnO_2 on the surface of Fe_3O_4.

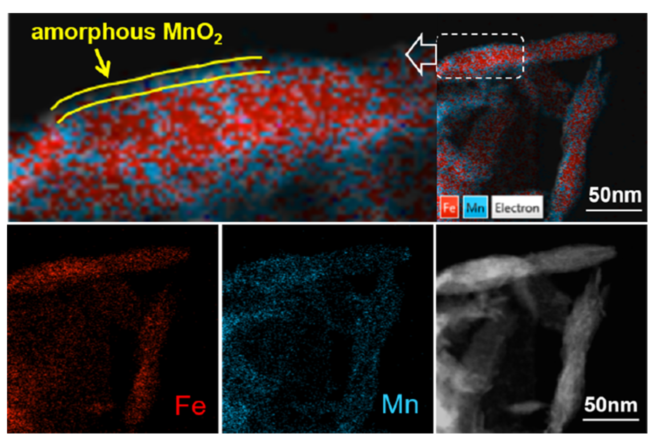

Figure 3. HRTEM image and EDX mapping images of fresh Mn-50.

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On the basis of the above characterization results, the preparation process of Fe_3O_4@MnO_2 nanoparticles with Fe_3O_4 particles encapsulated by amorphous MnO_2 is illustrated in Scheme S1. First, the ferrous sulfate solution turns to a green ferrous hydroxide suspension after the addition of sodium hydroxide. Subsequently, $KMnO_4$ added into the ferrous hydroxide is quickly reduced to MnO_2, accompanied by the formation of Fe_3O_4. Without the addition of $KMnO_4$, $Fe(OH)_2$ will be oxidized to FeOOH by the dissolved oxygen in the solution and then be continually converted to Fe_3O_4 nanoparticles via the dehydration of ferrous/ferric hydroxide.[30] The PVP added in the hydrothermal process strongly adsorbs on the surface of Fe_3O_4 particles, limiting crystal aggregation. The nucleated Fe_3O_4 nanoparticles can undergo further growth during synthesis time. Generally, anisotropic growth along the plane results in the formation of two-dimensional (2D) structural nanoplates.[28] The PVP surfactant in this experiment served as a linking agent by attaching to the lateral surface of the nanoparticles to form the 2D nanoplates. When the oxidant $KMnO_4$ is introduced, it is quickly reduced to MnO_2 and then adhered onto the surface of Fe_3O_4 nanoparticles via the enhanced affinity of hydroxyl radicals produced on the MnO_2 surface, generating a uniform nanometric shell. Furthermore, the decrease in Fe_3O_4(111) and/or (311) facet surface energy caused by MnO_2 facilitates anisotropic growth to large 2D nanostructures. Therefore, the increasing $KMnO_4$ content promotes the formation of more plate-shaped and flattened architectures of Fe_3O_4@MnO_2.

Nitrogen isotherms with the Barrett−Joyner−Halenda pore size distribution plots for all prepared samples are presented in Figure S2. All of the catalysts display the IV type isotherm with an apparent hysteresis loop on the basis of IUPAC classification.[31] A typical H1 hysteresis loop appears in the region above 0.9 p/p_0, which is assigned to a wide mesoporous size distribution,[32,33] suggesting that the prepared Fe_3O_4@MnO_2 has a shaggy structure. The pore size distribution presented in Figure S2 also shows similar mesopores for all prepared samples. The corresponding textural properties are summarized in Table 1. The BET surface area and average

Table 1. Textural Properties and Elemental Composition in the Bulk and the Surface of the Fresh Fe_3O_4@MnO_2 Catalysts

				Mn/Fe molar ratio		
catalysts	BET surface area ($m^2 \cdot g^{-1}$)	pore volume ($cm^3 \cdot g^{-1}$)	average pore size (nm)	bulk (from ICP)	surface (from XPS)	average crystallite size (nm)[a]
Mn-free	13.45	0.06	20.78			40.7
Mn-10	32.71	0.22	27.05	0.11	0.20	38.5
Mn-18	65.77	0.48	27.65	0.22	0.56	20.7
Mn-33	85.10	0.51	25.86	0.49	1.89	16.5
Mn-50	191.59	0.73	12.47	0.96	4.76	15.9

[a]Calculated from the Debye−Scherrer equation.

pore volume for Mn-free are 13.45 $m^2 \cdot g^{-1}$ and 0.06 $cm^3 \cdot g^{-1}$, respectively, which increase to 191.59 $m^2 \cdot g^{-1}$ and 0.73 $cm^3 \cdot g^{-1}$, respectively, for Mn-50, indicating that the incorporation of Mn species results in the excellent dispersion of Fe_3O_4 nanoparticles. Similar results were reported by Das et al., who showed that the addition of manganese to Fe catalysts supported on silicalite-1 could enhance the dispersion of iron oxides and reduce the particle size.[34] The average Fe_3O_4 crystalline diameter of Mn-free calculated by the Scherrer equation according to the XRD results is 40.7 nm, which gradually decreases to 15.9 nm as the Mn loading increases to 50% (Table 1), further confirming good dispersion and reduced particle size of iron oxide at higher Mn loading. In addition, an average pore diameter in the range from 12 to 27 nm was observed for all catalysts, verifying the mesoporous structure of Fe_3O_4@MnO_2 prepared by the one-pot hydrothermal method. In particular, the main porosity above 30 nm (shown in Figure S2) may be derived from the accumulation of the Fe_3O_4@MnO_2 elongated nanoparticles, which creates extra nanometric interparticle voids.[35,36] From Table 1, it is also found that the value of the Mn/Fe molar ratio is much higher in the surface layers than that in the bulk regions, demonstrating that Mn species accumulate on the surface and forms a nanometer-thick shell, which is consistent with the EDX mapping results.

The phase compositions in the bulk regions of the spent Fe_3O_4@MnO_2 catalysts are characterized by XRD (Figure S3). A new broad diffraction peak appears at a 2θ of approximately 43° corresponding to iron carbides (Fe_XC).[37] The intensity of the Fe_XC diffraction peaks decreases with the gradual increase of the Mn content, possibly because of the excellent dispersion of iron species at higher Mn loading. The surface compositions of fresh and spent catalysts are analyzed by XPS. Figure 4A shows that in the full-survey-scan spectrum, both the Fe 2p and O 1s signals are detected for the fresh single Fe_3O_4 catalyst, whereas an extra Mn 2p signal is observed for the fresh Fe_3O_4@MnO_2 catalysts. Figure 4B,C presents the detailed surface states of the Fe 2p and Mn 2p peaks, respectively. Fe $2p_{1/2}$ and Fe $2p_{3/2}$ peaks are observed at approximately 723.5 and 710.8 eV, respectively, with a shoulder peak at approximately 719 eV corresponding to the Fe_3O_4 phase.[38] The peak intensity of Fe shows a decreasing trend as the Mn content increases, which may be attributed to the formation of a thicker MnO_2 shell on the surface of Fe_3O_4, preventing the detection of surface Fe_3O_4. In particular, as the Mn content increases, the Fe 2p peak gradually shifts to a lower binding energy. A decrease of approximately 0.3 eV in binding energy is observed as the Mn content increases from 10 to 50%, which may be ascribed to the enhanced electron transfer from MnO_2 to Fe_3O_4 at higher Mn loading.[39] On the other hand, it is found that the Mn $2p_{3/2}$ and Mn $2p_{1/2}$ peaks appear at approximately 642.2 and 654.0 eV, respectively, corresponding to MnO_2 (Figure 4C). Compared to the Fe 2p peaks, the Mn 2p peaks shift toward a higher binding energy with the increasing Mn content, which is ascribed to the migration of electrons from MnO_2 into Fe_3O_4. Therefore, it is considered that the special structure of the Fe_3O_4 core coated by the MnO_2 shell enhances the electron transfer from MnO_2 to Fe_3O_4.[1] Moreover, with the addition of more Mn species, the intensity of the MnO_2 peaks on the surface layers exhibits an increasing trend, suggesting the formation of the MnO_2 shell. The XPS spectra of the spent catalysts shown in Figure S4 indicate that when compared to that for fresh catalysts, the Mn/Fe molar ratio in the surface layers for spent catalysts slightly decreases, implying that Fe migrates slowly to the surface layers during the reaction. Additionally, the signal intensity of both the Fe and Mn peaks decreases after the reaction. Only a faint intensity is detected for the Fe_3O_4@MnO_2 catalyst with higher Mn loading (such as Mn-33 or Mn-50), probably because of the production of more carbonaceous

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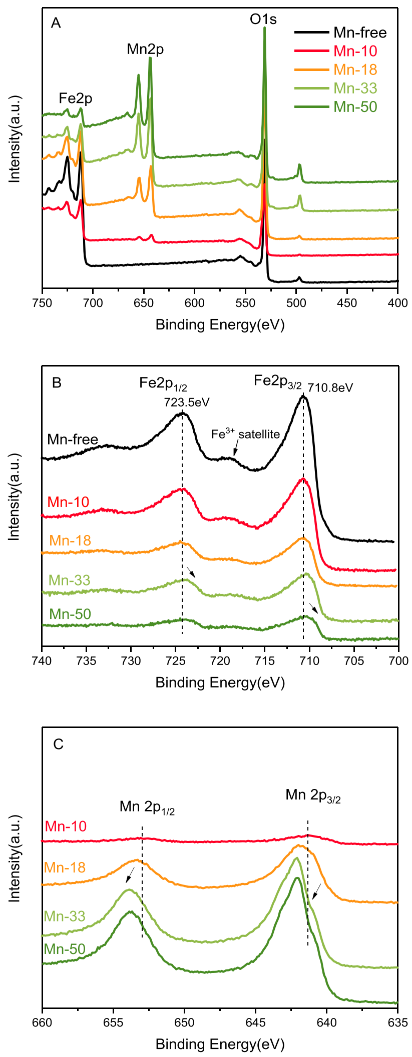

Figure 4. XPS spectra of the as-synthesized Fe_3O_4@MnO_2 catalyst (A) and high-resolution XPS spectra of Fe 2p (B) and Mn 2p (C).

species on the catalyst surface, limiting the detection of the surface Fe and Mn elements.

The surface metal oxides and carbonaceous species of the spent Fe_3O_4@MnO_2 catalysts are characterized by LRS. As shown in Figure 5, a broad band at around 700 cm^{-1} is observed for all spent catalysts, which can be assigned to the Fe_3O_4 phase. In addition, two main peaks observed at 1342 and 1607 cm^{-1} can be attributed to the disordered (D type) and ordered (G type) carbonaceous species, respectively.[40] Apparently, the intensity of the Fe_3O_4 peak decreases at higher Mn loading, probably because of the enhanced conversion of Fe_3O_4 to Fe_XC during the reaction with the catalysts having a higher Mn content. Furthermore, Mn enhances the dispersion of iron species and decreases the iron particle size as well, which may limit the detection of iron species by LRS. However, the intensity of both the D type and G type peaks presents a marked increasing trend as the Mn/Fe molar ratio increases, demonstrating that the incorporation of more Mn species into Fe_3O_4@MnO_2 facilitates the formation of carbonaceous species. Less Fe_3O_4 and more carbonaceous species are detected on the surface layers of the spent Fe_3O_4@MnO_2 catalyst with a higher Mn content, implying that the Mn promoter enhances the formation of Fe_XC, which is widely considered the active sites for FTS.[41−43]

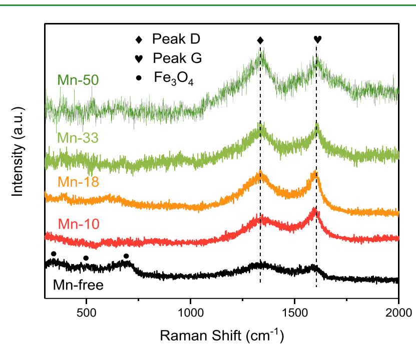

Figure 5. Raman spectrum profiles of the spent Fe_3O_4@MnO_2 catalyst.

The carbonaceous species of all spent catalysts are further analyzed by TGA (Figure S5). As can be seen, there are two weight-changing regions in the TGA pattern. The weight-gain region in the range from 200 to 350 °C may be attributed to the oxidation of Fe_3O_4 to Fe_2O_3 in air.[44] A wide weight-loss region appearing over 320 °C can be ascribed to the combustion of carbonaceous species on the surface.[45] Additionally, the amount of carbonaceous species of the spent catalysts increases with the increasing Mn content. The weight loss increases to 28 and 32% for Mn-33 and Mn-50, respectively, indicating that large amounts of carbonaceous species are formed on the Fe_3O_4@MnO_2 catalyst with higher Mn loading during the reaction.

The reduction behavior of the Fe_3O_4@MnO_2 catalysts with different Mn loadings is analyzed by H_2-TPR. As shown in Figure 6, two main reduction peaks appear in the regions of

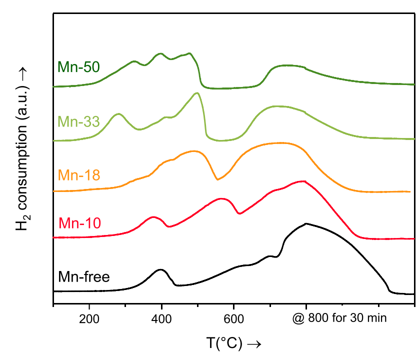

Figure 6. H_2-TPR profiles of the fresh Fe_3O_4@MnO_2 catalyst.

350−450 and 600−800 °C for the Mn-free catalyst, which are related to the continual reduction of $Fe_3O_4 \rightarrow FeO \rightarrow Fe$.[46] For Fe_3O_4@MnO_2 catalysts, in addition to the reduction peaks of iron oxides, a new main peak with a shoulder is detected at around 400−600 °C, corresponding to the reduction of MnO_2.[47] It is worth noting that the reduction peaks slightly shift to low temperature with the addition of Mn species, demonstrating that MnO_2-coated Fe_3O_4 nanoparticles are more easily reduced than that of Mn-free catalyst. This result is not contradictory to the previous results that showed that the added Mn species is incorporated into the lattice of iron, forming a Fe/Mn solid solution and suppressing the reduction of iron oxides because of the enhanced interaction of Fe/Mn.[48] In fact, this finding indicates that no Fe/Mn solid solution is formed in the Fe_3O_4@MnO_2 catalysts, which agrees well with

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Table 2. Catalytic Performance of Fe_3O_4@MnO_2 Catalysts in the FTS Reaction[a]

catalyst	activity [$mol_{CO} \cdot g_{Fe}^{-1} \cdot s^{-1}$]	CO conv. (%)	CO_2 sel. (%)	hydrocarbon sel. (%) CH_4	C_2-C_3	C_4^+	alkenes sel. (%) $C_2^{+=}$	$C_4^{+=}$	LAOs/O[b] (%)	α	C balance (%)
Mn-free	2.22×10^{-5}	87.64	47.61	15.06	28.83	56.10	45.58	32.65	51.08	0.68	93.94
Mn-10	2.52×10^{-5}	84.79	48.48	7.78	19.06	73.16	69.01	52.86	65.04	0.74	95.02
Mn-18	2.96×10^{-5}	81.37	47.60	9.89	25.25	64.86	59.76	39.43	49.01	0.76	94.95
Mn-33	3.96×10^{-5}	73.68	47.16	4.49	16.84	78.67	79.67	63.59	86.55	0.82	93.90
Mn-50	5.66×10^{-5}	67.90	47.11	3.58	14.63	81.79	79.34	65.81	87.10	0.83	92.35

[a]Reaction conditions: catalyst (0.5 g), syngas (CO/H_2 = 1:1), 3000 h^{-1}, 20 MPa, 280 °C. Pretreated with H_2 at 350 °C for 10 h. The data are collected at 30 h. [b]The ratio of linear α-$C_4^{+=}$/$C_4^{+=}$, LAOs stands for linear α-C_4^+ alkenes.

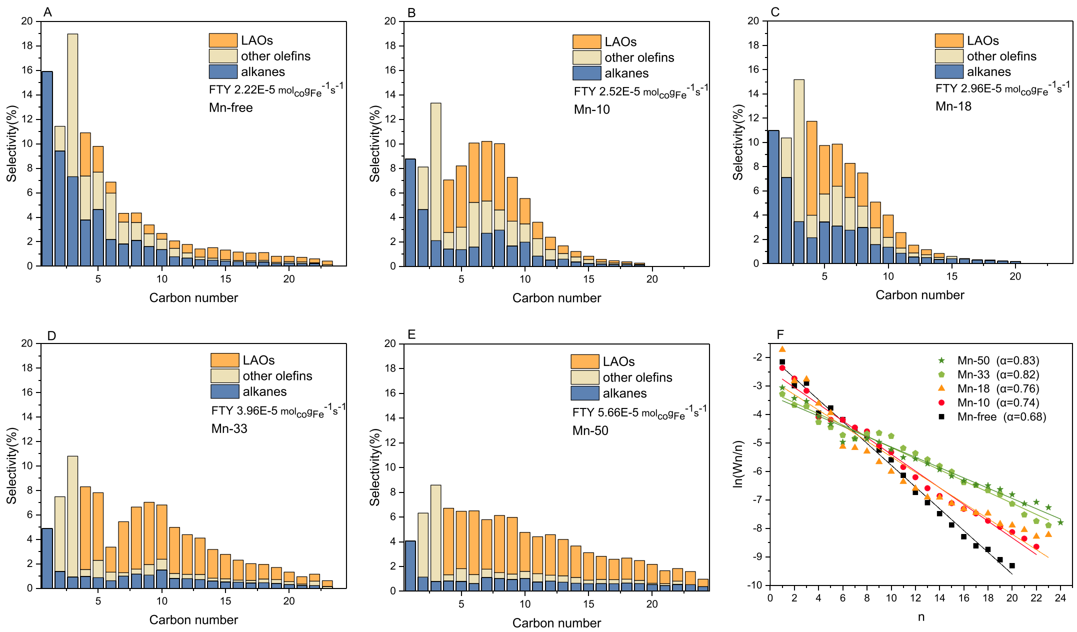

Figure 7. Product distribution of Fe_3O_4@MnO_2 catalysts with different Fe/Mn molar ratios (A–E) and ASF plots of these catalysts based on the product distribution after 30 h on stream (F) (the bars are stacked).

the TEM and XRD results. In addition, the enhanced electron transfer from MnO_2 to Fe_3O_4 in the Fe_3O_4@MnO_2 core–shell catalyst with higher Mn loading, as confirmed by XPS, promotes the reduction of iron oxides. As the amount of Mn species reaches 50%, the reduction peaks of iron oxides slightly shift to high temperature, which is mainly caused by the excessive covering of MnO_2 on the surface of Fe_3O_4, restraining the adsorption of hydrogen on Fe_3O_4 and decreasing the reduction of iron oxide. It reflects that excessive MnO_2 has a relatively intense interaction with Fe_3O_4, which is consistent with previous studies.[49,50]

3.2. Catalytic Evaluation of Fe_3O_4@MnO_2. The catalytic activity of Fe_3O_4@MnO_2 catalyst with different Mn loadings is displayed in Figure S6. To evaluate the conversion efficiency per gram of iron in unit time, FTY is chosen to estimate the catalytic activity. The activity (FTY) of single Fe_3O_4 is 2.22×10^{-5} $mol_{CO} \cdot g_{Fe}^{-1} \cdot s^{-1}$. The addition of the Mn promoter promotes the catalytic activity of the Fe_3O_4@MnO_2 catalyst. Compared to the Mn-free catalyst, the FTY value of the catalyst with 50% Mn promoter increases to 5.56×10^{-5} $mol_{CO} \cdot g_{Fe}^{-1} \cdot s^{-1}$, which is two times higher than that of the former. This result contrasts with previous reports in which the introduction of Mn species to iron-based catalysts decreased the catalytic activity owing to the strong interaction between Fe and Mn forming solid solutions.[51] Combined with the XRD and H_2-TPR results mentioned above, Mn species does not replace the octahedral site in Fe_3O_4. The increase in Mn loading results in the enhanced dispersion and smaller crystal size of iron species. Furthermore, the electron transfer from the MnO_2 shell to the Fe_3O_4 core is enhanced at higher Mn loading, enhancing the adsorption of CO on Fe species and the reduction of iron oxide, as indicated by XPS and H_2-TPR. As confirmed by XPS and LRS of the spent catalysts, a higher Mn content facilitates the formation of iron carbides, which are widely considered to be FTS active sites. Therefore, the Fe_3O_4@MnO_2 catalyst with higher Mn loading presents excellent catalytic activity. More interestingly, the FTY value of Fe_3O_4@MnO_2 is two times higher than the best value (2.70×10^{-5} $mol_{CO} \cdot g_{Fe}^{-1} \cdot s^{-1}$) of the MnO_2-coated Fe_2O_3 spindles reported by Zhang et al.,[27] suggesting that Fe_3O_4@MnO_2 in our work shows a better catalytic activity than the Fe_2O_3@MnO_2 spindles. It is possible that the higher BET surface area of Fe_3O_4@MnO_2 (~191 $m^2 \cdot g^{-1}$) than that of Fe_2O_3@MnO_2 (~10 $m^2 \cdot g^{-1}$) provides more iron active sites for the FTS.

In addition to the FTS activity, the product distribution of the Fe_3O_4@MnO_2 catalysts is shown in Table 2. It can be found that the CO_2 selectivity of the Mn-free catalyst is about 47%, which remains unchanged basically with the gradual addition of Mn species. The formation of CO_2 is mainly stemmed from the WGS reaction because of the excellent

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WGS ability of Fe-based catalysts, which is consistent with the results,[8,27] displaying a higher CO_2 selectivity above 40%. Besides, the Mn-free catalyst exhibits 15.06% CH_4, 28.83% C_2-C_3, and 56.10% C_4^+ hydrocarbons. The addition of the Mn species promotes a shift in the product distribution toward heavy hydrocarbons. As the Mn content increases to 50%, the selectivity of CH_4 and C_2-C_3 hydrocarbons decreases significantly to 3.58 and 14.63%, respectively, whereas that of C_4^+ hydrocarbons increases to 81.79%. In particular, the core−shell Fe_3O_4@MnO_2 catalyst facilitates the production of LAOs (Figure 7). The selectivity of C_4^+ alkenes increases from 32.65% for the Mn-free catalyst to 65.81% for the Mn-50 catalyst (Figure 7A−E), and the LAO selectivity in C_4^+ alkenes increases from 51.08 to 87.10%, demonstrating that the LAO products are easily formed with catalysts having high Mn loading. As the Mn loading increases to 60% (Mn-60, Table 2S), the C_4^+ alkene selectivity and LAO selectivity in C_4^+ alkenes decrease to 59.76 and 72.60%, respectively, although the FTY activity increases slightly to 6.11×10^{-5} $mol_{CO} \cdot g_{Fe}^{-1} \cdot s^{-1}$, indicating that the incorporation of excessive Mn results in a decrease of LAO selectivity. When the contact time over the Mn-free catalyst is adjusted to achieve the same FTY activity as with the Mn-50 (Table S2), the product distribution remains basically unchanged, suggesting that the Mn species has a crucial effect on the selectivity of alkenes and LAOs.

Generally, light olefins produced in the FTS reaction undergo an oligomerization reaction at the active sites to form long-chain LAOs. As the primary intermediates of C−C coupling, light olefins may also undergo a secondary hydrogenation reaction at the active sites. Therefore, enhancing the C−C coupling activity and suppressing the hydrogenation activity of iron-based catalysts can promote the production of LAOs. The XPS results indicate that the electron transfer from MnO_2 to Fe_3O_4 increases the amount of electrons on the surface of Fe species, which facilitates the adsorption of CO molecules and activates C−O bonds, enhancing the C−C coupling ability of the catalyst.[47] Moreover, the Mn-free catalyst presents a higher selectivity for CH_4 and C_2-C_3 hydrocarbons (Figure 7), which decreases as more Mn promoters are added, implying the spillover of dissociated H from iron onto manganese oxide. This spillover weakens the H adsorption at the active sites and decreases the hydrogenation ability of the catalyst.[52] On the other hand, it is generally accepted that the Fe/Mn solid solution facilitates the production of C_{2-4} olefins[53,54] and suppresses their carbon growth.[55] Zhang et al.[56] prepared a Fe/Mn spinel oxide, which exhibited a lower selectivity of C_5^+ hydrocarbons (42%) because it inhibited CO dissociation and carbon chain growth. Therefore, the absence of the Fe/Mn solid solution from the Fe_3O_4@MnO_2 core−shell catalyst in this study enhances the dissociation of C−O bonds and favors the chain growth of intermediate lower olefins, further improving the formation of LAOs. The ASF model is applied to predict the hydrocarbon product distribution. As shown in Figure 7F, the α value gradually increases with the increase in Mn loading. The α value of the Mn-free catalyst is 0.68, which increases to 0.83 as the Mn loading increases to 50%, indicating that the C−C coupling ability of the catalyst is well-enhanced with Mn addition. Particularly, a decreasing trend in the CH_4 selectivity was observed as the Mn content in the Fe_3O_4@MnO_2 catalyst increased, which indicates the inhibition of the hydrogenation reaction[27] and contributes to the enhanced formation of LAOs over the Fe_3O_4@MnO_2 core−shell catalyst with higher Mn loading.

3.3. Effect of the Reaction Conditions. In addition to the Mn content, the reaction conditions may also have a strong effect on the catalytic reaction performance. To maximize the production of LAOs, the optimal Mn-50 catalyst was selected to optimize the reaction conditions (including temperature, GHSV, and pressure). The effect of the reaction temperature is shown in Figure S7. CO conversion increases from 44.12 to 87.58% when the reaction temperature increases from 280 to 360 °C (Figure S7A). Simultaneously, C_4^+ hydrocarbon selectivity drops from 82.56 to 63.92%, whereas that of CH_4 and C_2-C_3 increases from 3.64 and 20.25 to 13.80 and 25.88%, respectively, indicating that a higher reaction temperature suppresses the C−C coupling process. Different from the impact described above, the C_4^+ alkene selectivity and LAO selectivity in C_4^+ increase with the increasing reaction temperature at first and then begin to decrease when the reaction temperature increases over 320 °C (Figure S7B). The optimal selectivity of LAOs in C_4^+ alkenes reaches 91.04% at 320 °C.

The impact of the reaction pressure on the catalytic evaluation of the Mn-50 catalyst at 320 °C and 3000 h^{-1} is shown in Figure S8. An increase in CO conversion from 57.13 to 91.03% with the increase in reaction pressure from 1.0 to 4.0 MPa (Figure S8A) demonstrates that a higher pressure promotes the catalytic activity of Mn-50 catalyst. The selectivity of total olefins and LAOs first slightly increases with the increasing reaction pressure and then begins to decrease with further increasing pressure (Figure S8B). A maximum value of 79.13% total olefins with 90.81% LAOs selectivity in C_4^+ alkenes is obtained at 2.0 MPa. The decrease in LAO selectivity at higher reaction pressures is possibly attributable to the increase in hydrogen partial pressure, promotion of the hydrogenation ability, and inhibition of the formation of LAOs.

Figure S9 shows the impact of GHSV on the catalytic performance of the Mn-50 catalyst. The CO conversion increases to 90.76% as GHSV increases to 6000 h^{-1} and then gradually decreases with the continual increase in GHSV. The elevated GHSV results in the product distribution shifting toward light hydrocarbons, and a similar trend is observed for the selectivity of total olefins and LAOs. On the basis of the reaction condition results, an optimized performance of syngas to LAOs is obtained over the Mn-50 catalyst at 2.0 MPa, 320 °C, and 3000 h^{-1}, which displays 75.47% CO conversion and an excellent selectivity of 79.60% total olefins and 64.95% C_4^+ alkenes, 91.04% of which are LAOs, as well as a low CH_4 selectivity of 5.24%. Compared to the other catalysts reported in Table S1, the Fe_3O_4@MnO_2 catalyst in this study exhibits an optimal catalytic performance for directly converting syngas to LAOs.

The optimized Mn-50 catalyst is used for the stability test. As shown in Figure 8, the Fe_3O_4@MnO_2 catalyst achieves a conversion of 50%, which basically remains unchanged during the reaction for 100 h. A lower selectivity of 4% CH_4 and 14% C_{2-3} hydrocarbons and a higher selectivity of 82% for C_4^+ hydrocarbons with 71% LAOs selectivity in total hydrocarbons are obtained during the reaction, indicating that the Fe_3O_4@MnO_2 catalyst presents not only excellent catalytic activity and LAO selectivity but also superior stability. According to the results of BET and XRD mentioned above, the incorporation of more Mn species into Fe_3O_4@MnO_2 results in the well-

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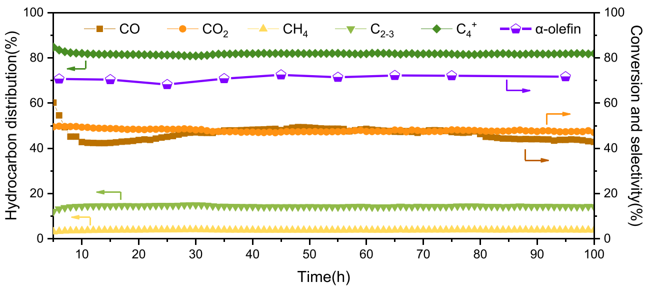

Figure 8. Stability of the Mn-50 catalyst. Reaction conditions: 280 °C, 2.0 MPa, and 6000 h^{-1} (α-olefin represents the percent of LAOs in all hydrocarbons formed).

dispersed iron oxide and smaller iron crystal size in the mesoporous core−shell structures. In addition, the TEM results shown in Figure S10 reveal that the iron nanoparticles remain in a stable state during the reaction. Therefore, the good dispersion and inhibition of the sintering of iron nanoparticles inside the core−shell structures play a significant role in improving the catalyst stability.[57] Moreover, the mesoporous structure of Fe_3O_4@MnO_2 facilitates the transfer of reactant and product molecules, further restraining catalyst deactivation. On the other hand, the initial Fe_3O_4 in the Fe_3O_4@MnO_2 catalyst can be converted more easily to iron carbides compared to the Fe_2O_3 phase, which readily increases the amount of active iron carbide sites, thereby enhancing the catalytic stability in the FTS reaction. By contrast, the Fe_2O_3@MnO_2 spindles prepared by Zhang et al. showed a decreasing trend in activity with time on stream, which could be attributed to the difficulty in the phase transformation from Fe_2O_3 to iron carbides, decreasing the active iron carbide sites for FTS.[27] In particular, the direct conversion of syngas to LAOs in one step simplifies the technological process compared to two steps (FTS + oligomerization), increasing the energy utilization efficiency. The novel Fe_3O_4@MnO_2 core−shell catalyst in this study displays a higher yield of LAOs and superior stability in comparison with the traditional Fe-based catalysts,[24,58] which improves the technology economy via the FTS route, providing a promising path for industrial applications.

4. CONCLUSIONS

In summary, a Fe_3O_4@MnO_2 core−shell catalyst was synthesized via a simple one-pot hydrothermal method, which realized the direct conversion of syngas to LAOs with high activity and selectivity. The catalyst displayed a selectivity of 79.60% total alkenes and 64.95% C_4^+ alkenes, in which LAOs accounted for 91%, with a much lower CH_4 selectivity of 5.24% at more than 75% CO conversion. The decrease in the Fe_3O_4 amount and increase in carbonaceous species amount on the surface layers at higher Mn loading facilitated the formation of more active iron carbides, improving the catalytic activity. The incorporation of more Mn species into Fe_3O_4@MnO_2 promoted the electron transfer from MnO_2 to Fe_3O_4, which facilitated the dissociative adsorption of CO molecules, detached H atoms from Fe species, enhanced C−C coupling, weakened the hydrogenation ability of active iron carbides, and improved the formation of LAOs. The mesoporous structure and excellent dispersion of iron species further strengthened the catalytic stability of Fe_3O_4@MnO_2.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acsami.8b11820.

Additional figures and tables as described in the text (PDF)

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